

Three moves to close the 0.044 gap on the table-tennis next-stroke leaderboard

The competitor's OOF-to-LB collapse is mostly a single problem in disguise: training on full rallies (mean=5.65) while test serves up truncated prefixes (mean=3.07). Reverse-engineering the LB score with the reported F1s implies test AUC $\approx$ 0.59 versus OOF AUC = 0.819 — a 0.23 collapse that alone explains $\sim$ 0.046 of leaderboard score, slightly more than the entire gap to first place. The three highest-ROI moves all hammer this same pressure point from different angles: (1) **random-prefix truncation augmentation matched to the test length distribution**, (2) **logit adjustment loss replacing the post-hoc additive-bias band-aid**, and (3) **cross-sport pretraining on ShuttleSet+ShuttleSet22 — schema-isomorphic badminton data with truncated-prefix evaluation built in**. The badminton route is uniquely de-risked because Hsu et al. (MJSSM 2026) literally publish the badminton $\rightarrow$ table-tennis transfer recipe with concrete numbers (Top-1 stroke 57.2%, landing 42.8%). Cross-sport pretraining is allowed by the rules and unexploited by virtually all competitors. (Mjssm + 2)

Below are the three ranked recommendations, ordered by expected lift per hour. Implement #1 first; it should be visible on a single LB submission and dictates whether #2 and #3 are worth pursuing.

#1 — Random-prefix truncation augmentation with length conditioning

Resource. The intellectual lineage is ASReP (Liu et al., SIGIR 2021, arXiv:2105.00522), BARec (arXiv:2112.06460, last revised Feb 2025, validated on $L \leq 3$ sessions), the 2024 sequential-recommendation augmentation survey (arXiv:2409.13545), and length-aware positional encoding (Takase, COLING 2020). None is sport-specific; the trick is a 5-line dataloader change. **The empirical pattern across these papers is unambiguous: random-prefix augmentation is the single largest lift on short-sequence test sets, typically 10–30% Recall@k.** No top-3 CoachAI Badminton Challenge 2023 team employed this — they trained on $\tau=4$ prefixes because the challenge fixed it — making it a genuinely uncrowded edge for this competition where τ varies and skews short.

Implementation (2–4h). In the train dataloader, for each rally of length L , sample a prefix length k from a distribution matching the empirical test histogram (concretely: roughly $P(k=1)=0.15$, $P(k=2)=0.25$, $P(k=3)=0.30$, $P(k=4)=0.20$, $P(k=5+)=0.10$ to hit mean 3.07). Train on (prefix_1..k, target_at_k+1). Add (step_idx=k) and (is_truncated=True) as both an LSTM positional embedding and a LightGBM feature. **Recompute every aggregate LightGBM feature over the prefix only** — this is critical because features like "rally length so far" or "count of forehand strokes" silently leak full-rally info otherwise, which is likely why the OOF-to-LB collapse is so severe. Re-tune the length-stratified plug-in bias on these new prefix-matched OOF predictions.

Expected lift and reasoning. OOF F1 will visibly drop (from 0.819 toward roughly 0.65–0.70 on AUC) because validation now matches the harder test regime. **LB should rise by 0.025–0.060**

composite score because the AUC component alone has ~0.04 of headroom recoverable, and macro-F1s will pick up another 0.01–0.02 as features stop overfitting to long-rally artifacts. This is the most lift-per-hour move in the entire research, because it fixes the diagnostic symptom (OOF/LB divergence of 0.463) directly.

1-hour kill criterion. After implementing the prefix-sampling dataloader and retraining the LSTM for one epoch, **check that OOF AUC on length-matched validation drops to roughly 0.65–0.75**. If it stays at 0.819, your "truncation" isn't actually being applied (most likely a feature still computed over the full rally). If it drops below 0.55, your prefix distribution is too aggressive — check the histogram match. If neither happens within 60 minutes of debugging, the LSTM architecture is leaking full-rally state through teacher-forcing or BiLSTM directionality; switch to causal LSTM and continue.

#2 — Logit-adjustment loss plus coordinate-ascent per-class bias

Resource. Menon et al., "Long-tail learning via logit adjustment," ICLR 2021 (arXiv:2007.07314, code at github.com/Chumsy0725/logit-adj-pytorch); Balanced Softmax (Ren et al., NeurIPS 2020, arXiv:2007.10740); and the December 2024 Prior2Posterior post-hoc correction (arXiv:2412.16540) which estimates the *effective* learned prior rather than the empirical class frequency. **Reported lift on long-tail benchmarks: post-hoc LA gives +3–7pp balanced accuracy on CIFAR-100-LT and ImageNet-LT; LA-loss adds another +1–3pp on top.** The user's mere +0.005 from post-hoc additive bias is below the typical range, which is itself a signal that the LSTM logits are head-saturated and need *training-time* correction, not output massaging.

[OpenReview](#) [arxiv](#)

Implementation (4h). Replace the LSTM softmax cross-entropy with $\text{F.cross_entropy}(\text{logits} + \tau * \log_prior, y)$ where $\log_prior = \text{torch.log}(\text{class_freq} / \text{class_freq.sum}())$ and $\tau \in \{0.5, 1.0, 1.5\}$ is grid-searched separately for action and point heads. **At inference, drop the adjustment** — predict from raw logits. For LightGBM, set `class_weight` to effective-number weights from Cui et al. (arXiv:1901.05555) with $\beta=0.999$: $w_c = (1-\beta)/(1-\beta^{n_c})$. Then replace the single-pass additive bias with **coordinate-ascent over 20 sweeps**: for each class, grid-search the additive bias offset over $[-3, 3]$ in 121 steps, holding others fixed; iterate. The user's current single-pass tuning leaves 2–4× of the achievable lift on the table per Kaggle macro-F1 folklore (LSHTC4 winners, Quora Insincere Questions #1). Add a P2P effective-prior correction in front of this for label-shift robustness. [Sanjivk](#) [arXiv](#)

Expected lift. **F1_action 0.371 → 0.395–0.410, F1_point 0.224 → 0.245–0.255**, composite +0.012 to +0.025. Stack additively with #1 because they fix orthogonal pathologies — LA fixes class imbalance in the loss landscape, prefix-truncation fixes input-distribution shift.

1-hour kill criterion. **OOF macro-F1 on the rarest 5 classes must improve by ≥30% relative** within one epoch of LA-loss training (e.g., a 0.10 → 0.13 jump on the worst tail). If tail F1 doesn't move, your `class_freq` vector is wrong (likely computed over a single fold rather than the global train distribution) or τ is mistuned — sweep τ once over $\{0.5, 1.0, 1.5, 2.0\}$ before abandoning. Total kill window is one epoch on a single fold — under 60 min on a 24GB GPU.

#3 — Cross-sport pretraining on ShuttleSet + ShuttleSet22

Resource. ShuttleSet (Wang et al., KDD 2023, arXiv:2306.04948) at github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet — 3,685 rallies / 36,492 strokes / 27 players, with **schema overlap of ~85% with the user's competition**: shot type (18 classes ≈ user's 19), landing area, player/opponent location, getpoint_player (≈ serverGetPoint), lose_reason (≈ point_id), backhand boolean, height, score. ShuttleSet22 (arXiv:2306.15664) at github.com/wywyWang/CoachAI-Projects/tree/main/CoachAI-Challenge-IJCAI2023 adds another 3,992 rallies / 33,612 strokes — and critically, **its val/test rallies are truncated to 4-stroke prefixes**, the closest public analog to the user's L=3.07 setting. Combined: ~70k strokes / ~7.7k rallies of pretraining data, comparable in scale to the user's 84,707 strokes. The decisive validation: **Hsu, Kondrič, Fan-Chiang & Wu (MJSSM 14(1):45–55, 2026, doi:10.26773/mjssm.260306) directly port ShuttleNet from badminton to table tennis on a private 2,225-rally Lin Yun-Ju dataset, achieving Top-1 stroke 57.2% and landing 42.8% — a published peer-reviewed proof of the exact transfer protocol.** (ADS + 8)

Implementation (6–8h). Clone the CoachAI-Projects repo, concatenate the two ShuttleSet CSV releases, and build a 22-class union vocabulary mapping the 18 badminton shot types to TT-compatible classes (net-shot/return-net/cross-court-net-shot → "short_push"; smash/wrist-smash → "smash"; lob/defensive-return-lob → "lob"; clear → drop; drive/back-court-drive/driven-flight → drive/fast-drive; drop/passive-drop → drop-shot; push/rush → push/flick; short/long-service → serve types). Map ShuttleSet's 9 in-court landing grids to the standard TT 9-zone grid; treat the 7 out-of-court grids as the "out" point_id proxy. Pretrain a bidirectional LSTM (hidden 128–256) with three heads — next-stroke 22-class CE, landing 9-zone CE, and "did server win the rally" BCE — using **the same prefix-truncation sampling from Recommendation #1**. Save the encoder. On the user's TT data, restore the encoder, reinitialize the three task-specific heads (since output cardinalities differ: 22→19, 9→10, BCE stays), reinitialize embedding rows for TT-only stroke classes (twist, chop, fast-push, pimples-push) to small Gaussian noise, and finetune with AdamW (lr=1e-4 body / 1e-3 heads, 5–10 epochs, prefix-sampling on). Extract LSTM hidden states as 128-d features for LightGBM. (arxiv)

Expected lift. +0.005 to +0.020 composite, concentrated on short-prefix predictions (k=1, k=2) where the user's TT-only model is most data-starved — exactly the regime the test set inhabits. Skip BadmintonDB (single-matchup, biases the model toward Momota/Ginting) and all video-only TT datasets (TTStroke-21, OpenTTGames, P2A, SPIN) — none have rally-level tabular sequences and feature extraction would consume the entire budget. Cite Wang et al. KDD '23 in the Kaggle external-data notebook. (arXiv)

1-hour kill criterion. After 1 epoch of badminton-only pretraining, **next-stroke top-1 on the ShuttleSet22 truncated val split must reach ≥0.30** (the published ShuttleNet baseline is around 0.30–0.35 on this split per the CoachAI Challenge 2023 leaderboard). If you stay at chance (~0.06 for 18 classes), the schema mapping is broken — most commonly the player/server identity got scrambled or the rally segmentation is misaligned. If pretraining works but the encoder doesn't transfer (TT finetune accuracy is no better than from-scratch), the embedding-row

reinitialization for TT-only classes likely overwrote shared semantics — try freezing the encoder for the first 2 finetune epochs.

Why this ordering and what to skip

Recommendation #1 is first because the OOF/LB divergence is mathematically the largest single error source and a 2-hour fix can't be matched by anything else in the research. **Run #1 alone on Day 1 and submit; expect 0.356 → 0.40+.** Recommendations #2 and #3 are stackable and orthogonal: #2 is faster but more incremental, #3 is slower but unique because external data is allowed and almost no competitor will have done schema-mapped cross-sport pretraining on truncated rallies. **If #1 underdelivers (LB lifts <0.015), abandon #3 and double down on #2** — the issue is then logit calibration, not distribution shift. If #1 overdelivers (LB jumps >0.04), invert priorities: #3 becomes the headroom play because every competitor will be converging on the calibration tricks of #2.

Skip the broader semi-supervised literature for this budget. FixMatch, FlexMatch, FreeMatch, SoftMatch, DASO, ABC, CoSSL, and SemiReward all assume image-style weak/strong augmentation pairs that don't exist for stroke-feature data — porting them costs more than the expected lift. The two tabular-validated SSL papers (R-CPL, arXiv:2302.14013; DIPS, arXiv:2406.13733) are worth bookmarking for a v2 push, but their reported lifts (+0.002 to +0.012) are dominated by #1-#3. Likewise skip Mamba (no published rally model as of May 2026), Diffusion Forcing (arXiv:2407.01392, NeurIPS 2024 — heavy implementation), TemPr-style multi-tower early-action (arXiv:2204.13340 — major refactor), and full ShuttleNet replacement (only worth it if you have spare budget after #1-#3 land). The hierarchical action head (already on the competitor's parallel track) and stacking meta-learner are sufficient for the 19-class taxonomy without adding a separate decomposition layer. [arxiv](#) [arXiv](#)

Bottom line

The leaderboard gap is one structural problem (truncation-induced distribution shift) plus one tactical problem (head-saturated logits) plus one untapped resource (badminton data with matching schema). Three commits — a dataloader change, a loss-function swap, and a pretrained-encoder load — should close it. **Submit #1 first; the kill criteria above will tell you within 60 minutes whether the path is real.**